

Accessibility Meets Neurodiversity : Designing and Developing with Empathy and Inclusion

Vaishnavi Venkata Subramanian



KA-CHOW!

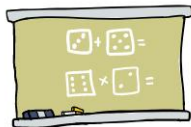
Hello,
I'm Vaishnavi



Vancouver



Things I ❤️



Teaching



Public Speaking



Visual Communication

Things I do 🧐



Software Engineering



Ideation



Team collaboration

How to Give Me Feedback 🧐

Direct constructive feedback provided with utmost honesty to help me grow.
Data driven and candid conversations with potential solutions for me to thrive and succeed.

Comms Preferences 🧑



Working Patterns 🌞



- Morning Person
- Enjoy a quiet and vibrant environment
- Follow Pomodoro Technique 🍅

Things I struggle with 🧑



Mickey Mouse Job
#halfbaked



Yak Shaving



Interruptions



“Simple things should be simple,
complex things should be possible.” - Alan Kay

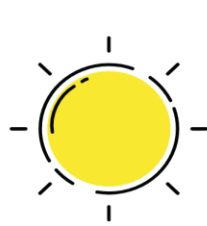


P.S: No animals or disney characters were harmed in the making of this manual 🐼 🐼 🐼



April 2023
Version 3

HOW YOU DOIN'



HAPPY



OKAY



MEH



WHERE'S MY
COFFEE



YAWN



SNOOZE

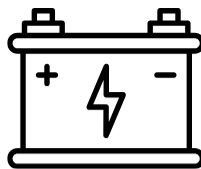


THIRSTY



HUNGRY

MORNING



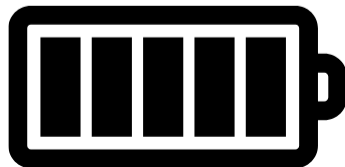
tired.

AFTERNOON



ready to nap.

BEDTIME



wide awake.

Before November 30, 2022
and
After



This presentation is for everyone!



Agenda

Accessibility vs Neurodiversity

Disabilities

Guidelines for Design

A few best practices

Concluding Remarks

SOME DISABILITIES LOOK LIKE THIS



OTHERS LOOK LIKE THIS



NOT ALL DISABILITIES ARE VISIBLE

Source : <https://greatcareers.org/neurodiversity-in-the-workplace-and-international-day-of-persons-with-disabilities/>

Permanent Temporary Situational

Touch



One arm



Arm injury



New parent

See



Blind



Cataract



Distracted driver

Permanent Temporary Situational

Hear



Deaf



Ear infection



Bartender

Speak



Non-verbal



Laryngitis



Heavy accent

Kinds of Disability



Permanent disability

Never goes away and it affects a person's daily life



Temporary impairment

Eventually goes away



Situational impairment

Brief as once the situation changes, it goes away



Episodic impairment

Comes and goes at unpredictable times and lasts for different amounts of time



Assistive Technologies

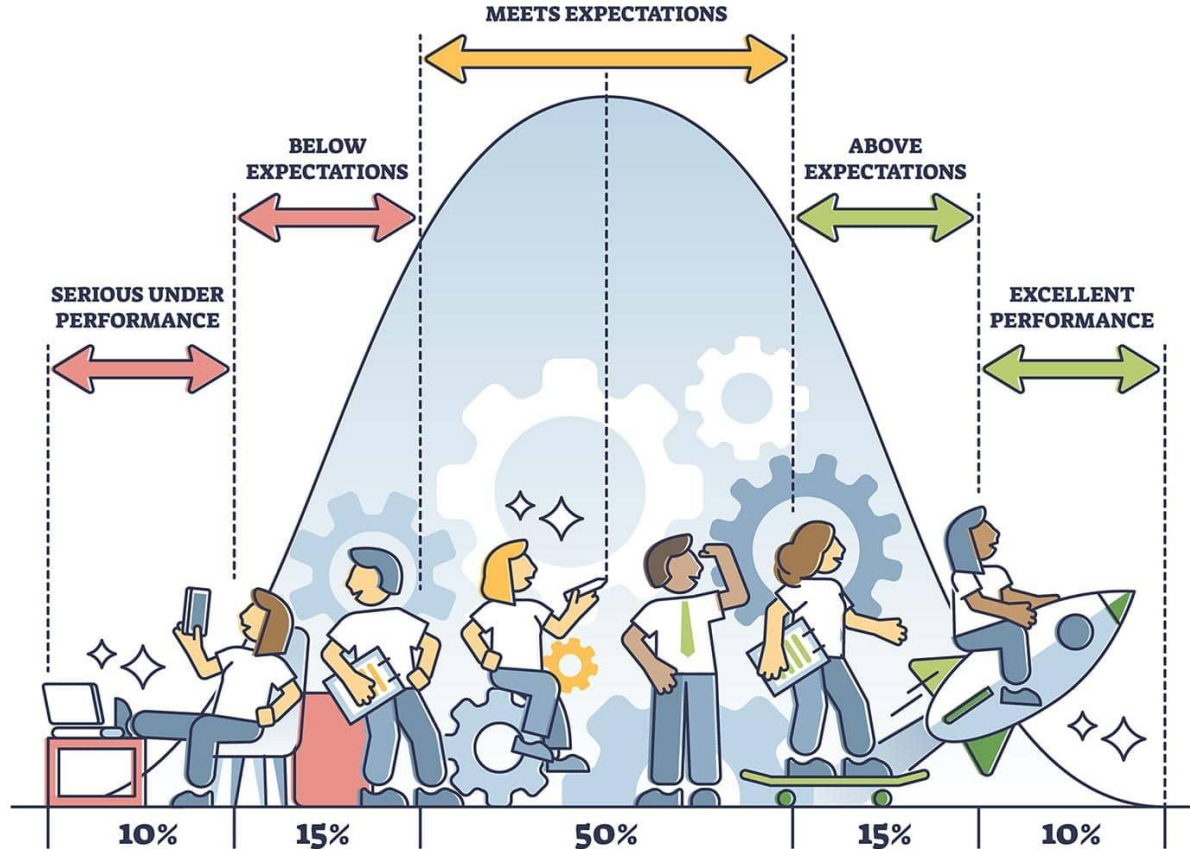
THE CURB-CUT EFFECT

When we design for disabilities,
we make things better for everyone

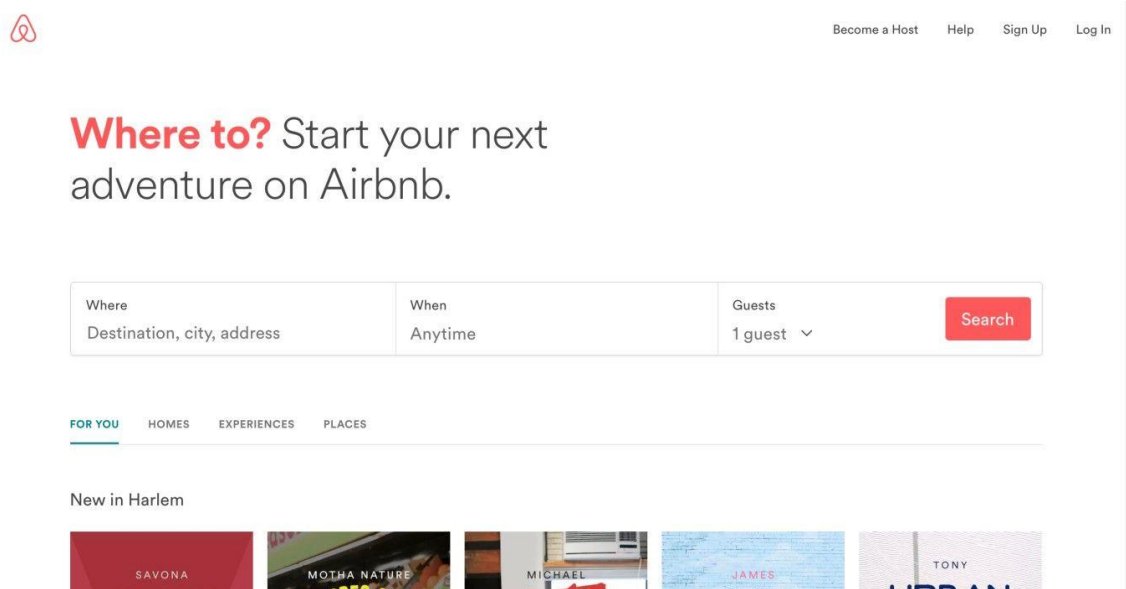




BELL CURVE



Implement a Forgiving Format to Accommodate Users' Mistakes



The image shows the Airbnb homepage with a search bar that has three input fields: 'Where' (with placeholder text 'Destination, city, address'), 'When' (with placeholder text 'Anytime'), and 'Guests' (with a dropdown menu showing '1 guest'). A red 'Search' button is located to the right of the 'Guests' field. Above the search bar, the Airbnb logo is on the left, and links for 'Become a Host', 'Help', 'Sign Up', and 'Log In' are on the right. Below the search bar, there is a section titled 'FOR YOU' with sub-sections 'HOMES', 'EXPERIENCES', and 'PLACES'. Under 'HOMES', there is a section 'New in Harlem' with five featured listings: 'SAVONA', 'MOTHA NATURE', 'MICHAEL', 'JAMES', and 'TONY'.

 [Become a Host](#) [Help](#) [Sign Up](#) [Log In](#)

Where to? Start your next adventure on Airbnb.

Where
Destination, city, address

When
Anytime

Guests
1 guest ▾

Search

FOR YOU [HOMES](#) [EXPERIENCES](#) [PLACES](#)

New in Harlem

SAVONA

MOTHA NATURE

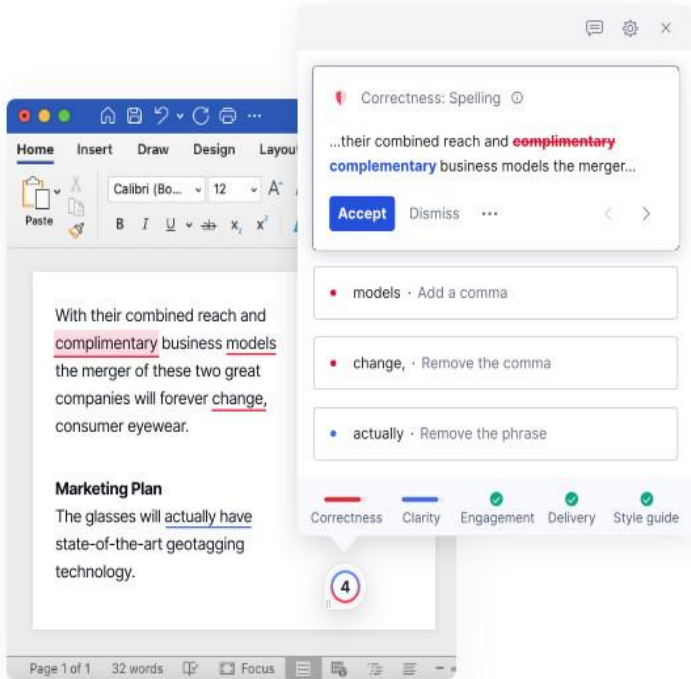
MICHAEL

JAMES

TONY

Source : <https://www.interaction-design.org/literature/article/how-to-implement-a-forgiving-format-to-accommodate-users-mistakes>

Be kind and forgiving



Grammarly - Disneyland of Spelling

Forgot attachment?

*My email that is well
written, reviewed and
ready to be sent*



Email sent



My attached files



Forgiveness



Search Mail

Search the Web

The conversation has been moved to the Trash. [Learn more](#) [Undo](#)

[Gmail](#) [Calendar](#) [Documents](#) [Reader](#) [Web](#) [more ▼](#)

Saving...

Google docs Sample document

saved on 11/16/09 3:35 PM by Janko Jovanovic

Create Abundance;
Not redundancy



Show less

Favorites

Recruiting weekly mee...

Website Launch

Rebranding video

Marketing

Editorial Calendar

Customer Stories - Q4

Website Design Reque...

Show more

Reports

Teams

Marketing



Customer Stories - Q4



On Track



Share

Search



List

Board

Timeline

Calendar

Dashboard

More...

Last task completed on 12 Oct

Incomplete tasks

Filter

Sort

Rules

Apps

Fields

...

Recurrent pieces



New formats



Content



Design

✓ Edit customer testimonial



19 Nov

1

✓ Customer Stories - Q4 submission



18 Nov

Content marketing campaign!



16 Dec

Editorial calendar



22 Dec

Create campaign



✓ Work-life balance newsletter



24 Nov

✓ Apollo for nonprofits



14 Jan, 2021

2

Quarter in review



25 Nov

2

Press release on acquisition



23 Dec

1 4

✓ Add testimonials



22 Dec

new



Monday

Changes



30 Dec

✓ Rewrite blog post



19 Jan, 2021

✓ Choose customer for February spotlight



20 Jan, 2021

Editorial calendar



29 Dec

✓ Customer spotlight



18 Jan, 2021

Create new in



17 Dec

My Tasks

Inbox

Dashboard

Bug tracking 1

Candidate tracking

Recruiting Pipeline

Recruiting weekly meeting

Editorial Calendar

Website Launch

Multihomed projects

Website Launch 2.0

Editorial calendar

Show recent projects

Reports

Teams

Everyone at Apollo Enter...

Marketing

Design

Managers

Recruiting

HR

Bug tracking



11



Share

Go to any project or task...



List

Timeline

Calendar

Conversations

Progress

Files

August 2018



Today



Mon

Tue

Wed

Thu

Fri

Sat

Sun

6

7

8

9

10

11

12



✓ Broken links on help page

✓ Search
is not...

13

14

15

16

17

18

19



✓ Update cookies notice



layout format is out off



20

21

22

23

24

25

26

Monthly Triage



Triage iOS bugs

✓ Log in
button...Daily
triage

Universal Design Foods (UDF)

Nursery to Nursing Homes

Category	Easy to chew	Can be chewed with gums	Can be broken down with tongue	Does not have to be chewed
Ability to chew	Somewhat difficult to eat hard or large pieces of food	Difficult to eat hard or large pieces of food	Can eat soft, small pieces of food	Cannot eat solid food, even if they are small pieces
Ability to swallow	Can swallow normally	Some foods are difficult to swallow	Beverages are sometimes difficult to swallow	Beverages are difficult to swallow
Softness (rice)	Between regular rice and soft rice 	Between soft rice and rice porridge 	Rice porridge 	Mashed rice porridge 
Softness (eggs)	Thick Japanese omelet 	Rolled Japanese omelet 	Scrambled eggs 	Soft steamed egg custard (no solid food ingredients) 

Canary in the coalmine



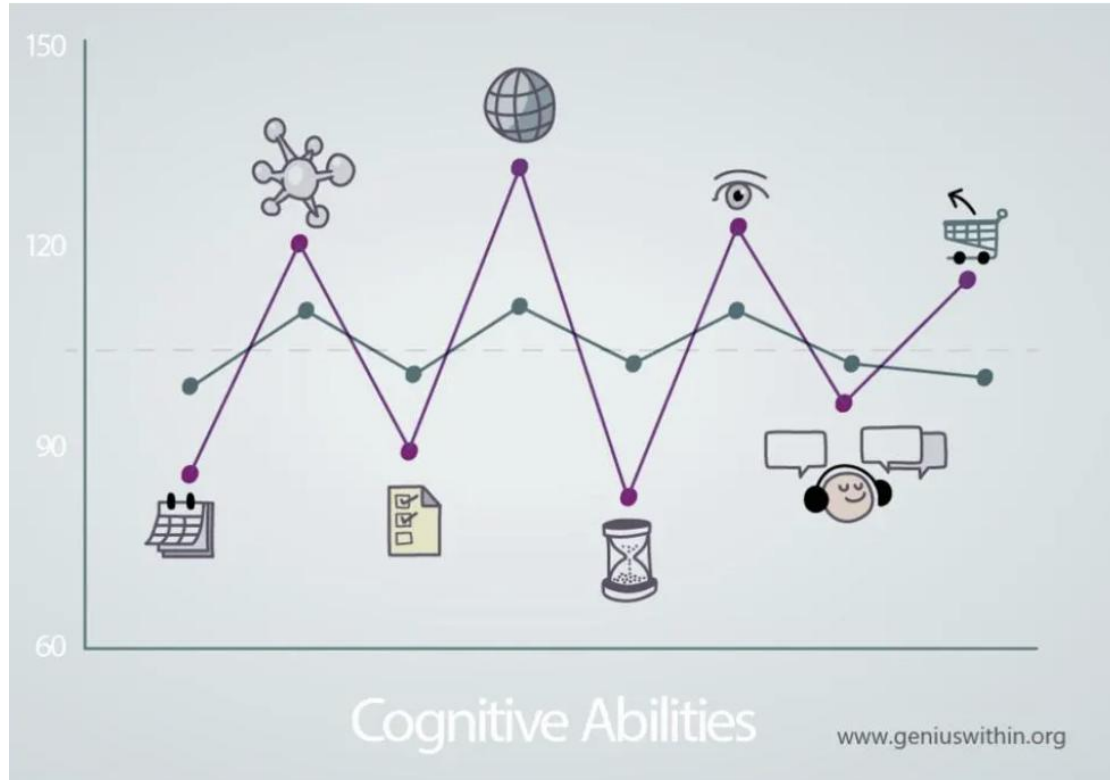
Often the first to sense danger akin to canaries in the coal mine



Preventing miscommunication and not overwhelming the users



Spiky Profiles



Guidelines

Guidelines

Interpretable messages :

Avoid use of characters that look alike

E.g. 1ll, B8, QO0CD, and break up long strings in phone numbers e.g. (000) 000–0000

Stick to active voice :

Passive voice sentences are longer and harder to comprehend.

Users with ADHD will get bored, dyslexics will have even more trouble reading, and so will readers on the spectrum.

Avoid metaphors :

Trouble understanding metaphors, such as idioms and other abstract figures of speech.

E.g. Instead of calling it a day, it's more inclusive to say that something is done.

Guidelines

Use emojis wisely :

Different languages have different interpretations. Custom and culture vary

Format your writing :

Divide large chunks of text into smaller paragraphs.

It is challenging for people struggling with limited attention spans.

Use descriptive buttons :

No vague microcopy, such as *Click here*. Think of screen readers

Describe the purpose of the button.

Example: *Upload images, Register, Download the e-book.*

This simple change makes navigation smoother and reduces anxiety.

Guidelines

Be careful with animations :

Too much stimulation can induce anxiety in people on the autism spectrum, which may even lead to physical pain, withdrawal, or meltdown.

Use of animated elements :

Should be limited to instances when it serves a purpose.
Also remember not to make them too invasive.

Mind the abbreviations :

AI in context

Artificial Intelligence vs Artificial Insemination

Guidelines

Avoid autoplay

Unexpected movement is surprising and scary,
not to mention the sound!

Psstt I startle a lot!

Add subtitles and captions

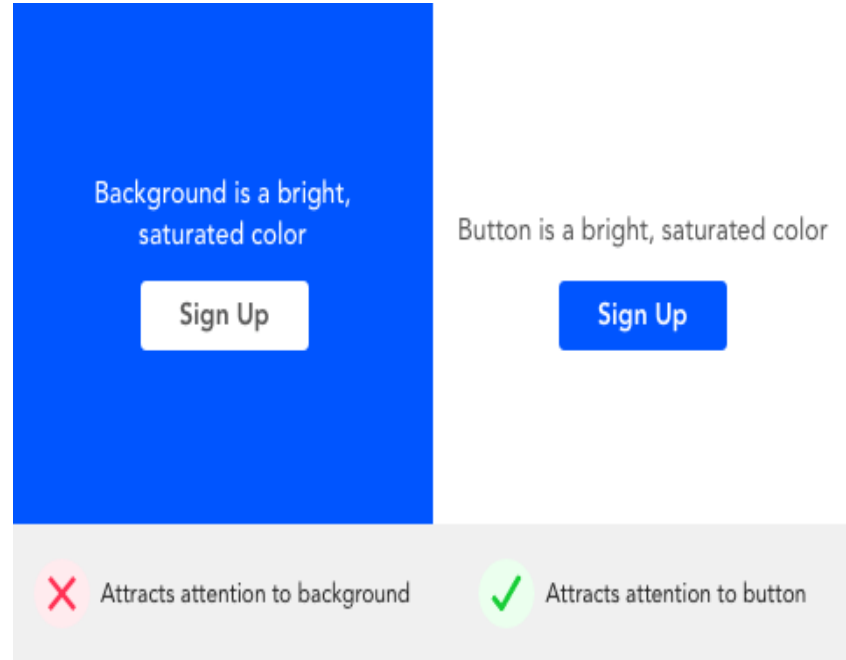


Aim to visually please, rather than disturb.

Abstain from auto play in websites and music that suddenly appears out of nowhere.

Skip highly vibrant colors.

Perform user testing and focus groups with ND users.



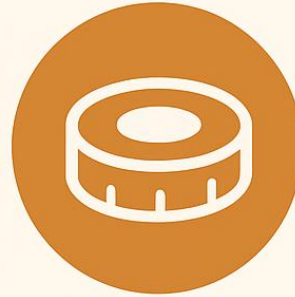
Clear instructions use...



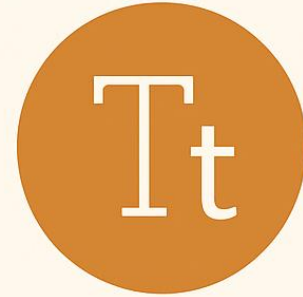
Short
sentences



Precise
terminology



Exact
measurement



Nouns, not
pronouns

Protanopia - Colour Blindness



Airport parking garage at PDX. Move slider to the left to see what Andy Baio sees.



Move slider to the left to see what Andy Baio sees. Image by PopSugar

Test in Grayscale. Mute the colors

Source : [colorblindness-design-ui-accessibility-wordle](https://colorblindness-design-ui-accessibility-wordle.com/)

Designing for users on the autistic spectrum



Do...

use simple colours



write in plain English

Do this.

use simple sentences and bullets



make buttons descriptive

Attach files

build simple and consistent layouts



Don't...

use bright contrasting colours



use figures of speech and idioms



create a wall of text



make buttons vague and unpredictable

Click here!

build complex and cluttered layouts



Designing for users with dyslexia



Do...

use images and diagrams to support text



align text to the left and keep a consistent layout



consider producing materials in other formats (for example, audio or video)



keep content short, clear and simple



let users change the contrast between background and text



Don't...

use large blocks of heavy text



underline words, use italics or write in capitals

DON'T
DO THIS

force users to remember things from previous pages - give reminders and prompts



rely on accurate spelling - use autocorrect or provide suggestions



put too much information in one place



DIMINISHING VISUAL GLARE WHEN DESIGNING COURSE MATERIAL FOR

NEURO-DIVERGENT KIDS



Use a neutral colored background.



Employ lower intensity of black color.



Separate sections or bullets by colors.



Reduce the intensity of screen brightness.



Avoid neon or bright hues, use natural tones.



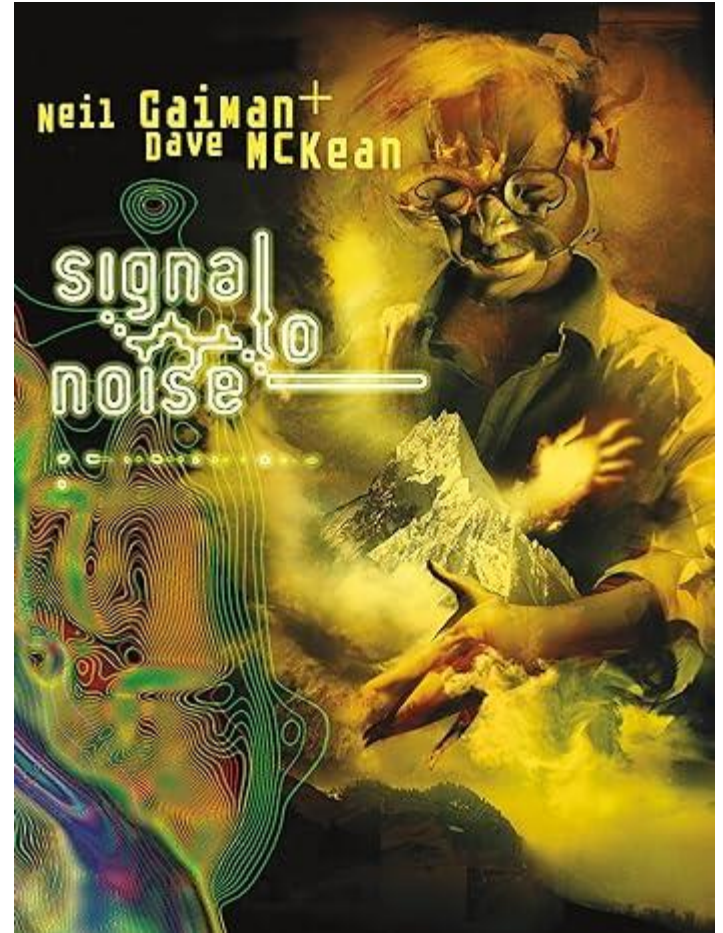
Use contrast to highlight call-to-action.

S/R (dB)



Aim for higher SNR

More useful information (the
signal) >>>>>>
unwanted data (the noise)



A design with inclusivity
and neurodiversity in
mind is not just 'nice to
have' – it's imperative



Accessibility is more than a ramp

Designing for 'disability first' often uncovers solutions that help everyone.

Great Minds Think
Unlike And Learn
Differently

Representation
Matters!



Nothing About Us Without Us – Empowering All Voices



Embrace
diversity,
celebrate each
unique story

Together, we can create a
brighter future.

**Inclusivity Starts With You and
Me**



References

- **Abundance** : <https://designingforneurodiversity.substack.com/p/abundance>
- **Forgiving Format Design Pattern** <https://www.interaction-design.org/literature/article/how-to-implement-a-forgiving-format-to-accommodate-users-mistakes>
- **UDF** : <https://www.kewpie.com/en/sustainability/dietary-lifestyle/universal/>
- [8 Everyday Items Originally Invented for People With Disabilities](#)

Thank you

